4/29 CptS 350

Final Exam Review II.

Re-do standard problems in hw:

Don't make mistakes on simple problems!

Classic algs, for instance,

Greedy Algs, Divide-Conquer, Dynamic Programing,

Names: Tarjan, Dijskstra, S. Cook, R. Karp.

Rivest, ...

Look at titles of lectures on canvas "Pages".

Ex: 4/6. RSA.

- what is it?
- what's the first step?
  second
- Is RSA absolutely secure?

3/11. SCC Alg.

- what is it? def?
- Roughly, what is the alg?
- Time complexity?

App. of SCC ——— cycles, ω-walks in a graph. q_{inf}.

Example: Design a linear-time alg to decide:

Given: a graph G

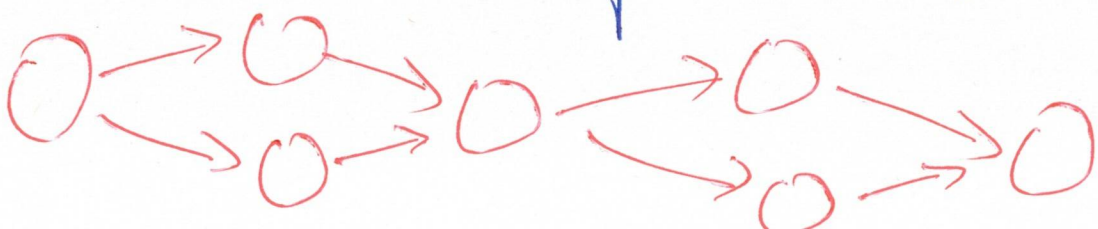
Question: Is G connected?

Run SCC on G and make sure that all nodes in G form one SCC.

Design a linear-time alg to decide:

Given: a graph G

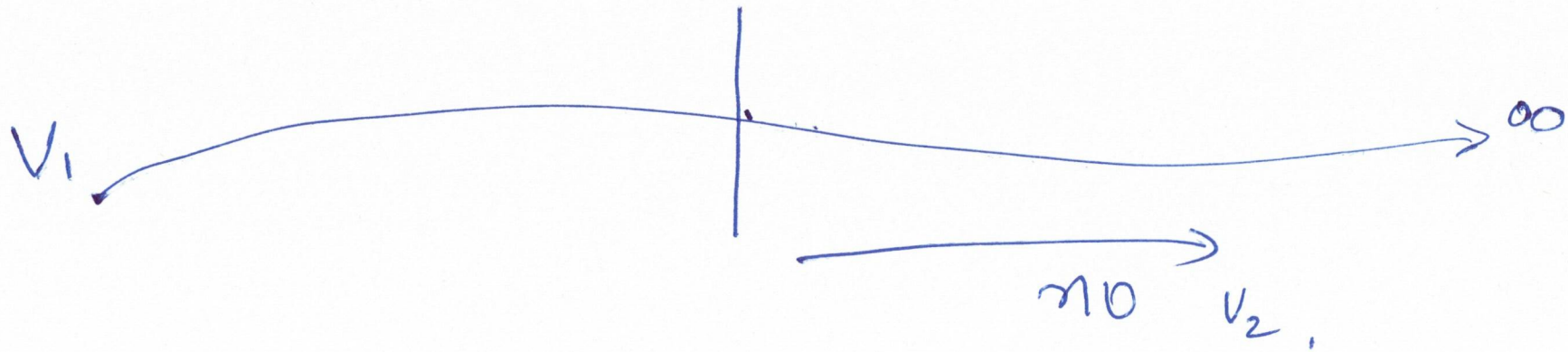
Question: Is G a DAG?



Run SCC and make sure that each SCC contains exactly one node (there is no self-loop on each node).

Given: a graph G with two designated nodes v_1, v_2 (G has many other nodes).

Qstion: Is there an inf-walk from v_1 s.t. it passes v_2 for only finitely many times?



Key: Drop v_2 in certain way.

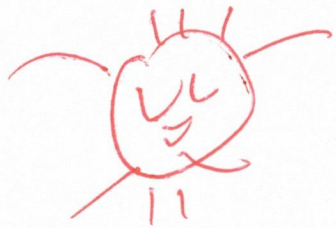


Alg: Run SCC on G with v_2 dropped.
if there no looping SCC, ret NO;
For each looping SCC, check if there is a walk in G from v_1 to a node in the SCC, (if yes), ret yes
Ret NO.

Given: a graph G with two designated nodes v_1, v_2.

Claim: Is there an inf walk from v_1 s.t. v_2 is passed for inf. many times and v_1 is passed for finitely many times?

(look at class notes under "Pages").



do it tonight.

Dynamic Programing:

Can you implement Fib(·) in linear time?

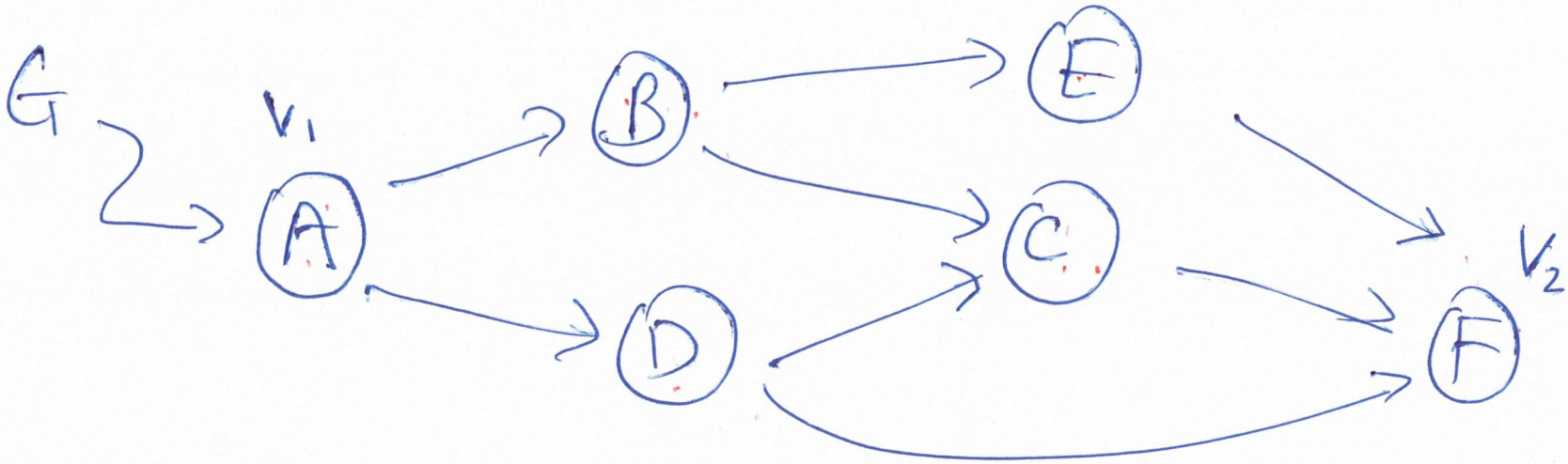
(use forward propagation) (pseudo-code)

Path counting on DAG.

Given: a DAG G with two designated nodes v_1 and v_2

Want: the total # of (distinct) walks from v_1 to v_2 in G.

(use pseudo-code or sth close to pseudo)



Run topo-sort(G) // call topo-sort().

we have:

	A	B	D	E	C	F
Forward-pop:	1	1	1	1	~~1~~ 2	~~1~~ 4
	1	∅	∅	∅	∅	∅
	↑	↑	↑	↑	↑	↑

✓

HW Prob:

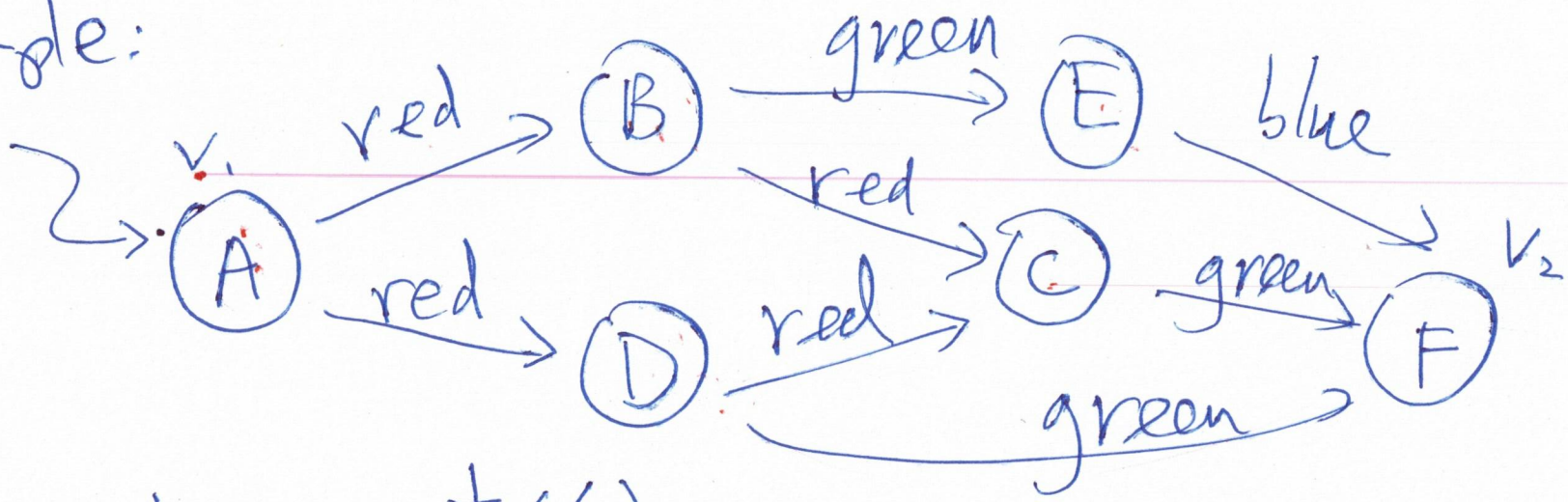
Given: DAG G with two designated nodes v_1, v_2. Each edge has a color.

Want: the total # of (distinct) walks from v_1 to v_2 in G s.t. each walk contains more red's than green's.

(pseudo-code).

look at class notes. (3/30).

Exmple:



Run topo-sort(G);

